

Chia-Hsiang Kao

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Ithaca, New York (14850)

Vision and Interest

My research vision is to build systems that infer and learn the hidden dynamics of real-world interactions. Specifically, I am interested in **Vision Reasoning**, **Visual-Symbolic Learning (image/video-to-code)**, **Video Editing**, and **AI Agent**.

I am currently a **Cornell Foundational AI PhD Fellow (2025)**, a Meta Student Researcher, and was a Applied Scientist Intern at **Amazon**.

Education

Cornell University, Ithaca, NY

2023 - Present

- Ph.D. Student in Computer Science
- Advisor: Professor [Bharath Hariharan](#)

National Yang-Ming Chiao-Tung University, Taipei, Taiwan

2015 - 2022

- Doctor of Medicine

Work Experience

Applied Scientist, Amazon; Sunnyvale, CA

May 2025 - Aug. 2025

- I developed VLMs that fully reconstructs the real-world rigid-body dynamics in MuJoCo simulator.
- We leveraged language as a structural visual representation and enhanced performance with event reasoning.

Student Researcher (part-time), Meta; remote

Dec 2025 - May. 2026

- I will develop video-to-video editing system for controlling and changing the real-world physics and motion in the video.

Publications

[U.1] **Inferring Physics from Videos for Rigid Object Motion Capture and Editing**

[Chia-Hsiang Kao](#), Chien-Yi Wang, Cong Phuoc Huynh, Bharath Hariharan

Submitted to CVPR 2026

[S.2] **Towards LLM Agents for Earth Observation.**

[Chia-Hsiang Kao](#), Wenting Zhao, Shreelekha Revankar, Samuel Speas, Snehal Bhagat, Rajeev Datta, Cheng Perng Phoo, Utkarsh Mall, Carl Vondrick, Kavita Bala, Bharath Hariharan

ICML 2025 workshop

[C.6] **Counter-Current Learning: A Biologically Plausible Dual Network Approach for Deep Learning**

[Chia-Hsiang Kao](#), Bharath Hariharan

NeurIPS 2024

[C.5] **AllClear: A Comprehensive Dataset and Benchmark for Cloud Removal in Satellite Imagery**

[Chia-Hsiang Kao](#)*, Hangyu Zhou*, Cheng Perng Phoo, Utkarsh Mall, Bharath Hariharan, Kavita Bala

NeurIPS Datasets and Benchmarks Track 2024

[C.4] **Caduceus: Bi-Directional Equivariant Long-Range DNA Sequence Modeling**

Yair Schiff, [Chia-Hsiang Kao](#), Aaron Gokalsan, Tri Dao, Albert Gu, Volodymyr Kuleshov

ICML 2024

[C.3] **Advancing DNA Language Models: The Genomics Long-Range Benchmark**

Chia-Hsiang Kao*, Evan Trop*, McKinley Polen*, Yair Schiff*, Bernardo P. de Almeida, Aaron Gokaslan,
Thomas Pierrot, Volodymyr Kuleshov
AAAI (workshop) 2023

[S.1] **FedBug: A Bottom-Up Gradual Unfreezing Framework for Federated Learning**

Chia-Hsiang Kao, Yu-Chiang Frank Wang
Preprint

[C.2] **MAML Is a Noisy Contrastive Learner in the Classification**

Chia-Hsiang Kao, Wei-Chen Chiu, Pin-Yu Chen
ICLR 2022

[C.1] **Demystifying T1-MRI to FDG18-PET Image Translation via Representational Similarity**

Chia-Hsiang Kao, Yong-Sheng Chen, Li-Fen Chen, Wei-Chen Chiu
MICCAI 2021

Honors & Awards

- **Foundational AI PhD Fellowship, Cornell** 2025
- Student Travel Award, *MICCAI* 2021
- Undergraduate Research Fellowship, *National Science and Technology Council, Taiwan* 2018, 2020
- Summer Research Fellowship, *National Health Research Institutes* 2018

Services

Conference Reviewer: AAAI[25], AISTATS[25], AutoML[22], ICML[25], ICLR[25], NeurIPS[21,24,25], ICLR [25, 26]

Journal Reviewer: CVIU (2022), Comput. Electr. Eng. (2024), IEEE TETCI (2024)